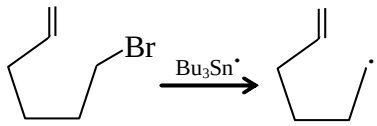
CHEMISTRY

SECTION – I
Straight Objective Type

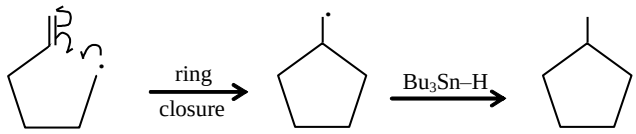
1. **(d)** Tributyl stannane reduces alkyl halide to corresponding alkane by free-radical mechanism.

CIS: $Me_2C(CN)–N=N–C(CN)Me_2 \xrightarrow{\Delta} N_2\uparrow + Me_2\dot{C}(CN)$

$Me_2\dot{C}(CN) + Bu_3Sn–H \longrightarrow Bu_3S\dot{n} + Me_2CH(CN)$

CPS: 

$+ Bu_3Sn–Br$



$+ Bu_3S\dot{n}$

Ease of formation of a five membered ring is greater than that of a six membered ring.

2. **(b)** $-\frac{d[A]}{dt} = \frac{k_1[A]}{1+k_2[A]}$

At high concentration of A ([A] >> k_2), $1 + k_2[A] \simeq k_2[A]$

$-\frac{d[A]}{dt} = \frac{k_1}{k_2} = k.$

Thus, order of the reaction is zero.

At low concentration of A([A] << k_2), $1 + k_2[A] = 1$

$-\frac{d[A]}{dt} = \frac{k_1[A]}{1} = k_1[A]$

The order of the reaction is unity.

3. **(a)** $n = \frac{N}{N_A} = \frac{1.004\times10^{23}}{6.023\times10^{23}} = \frac{1}{6}.$

$T = \frac{PV}{nR} = \frac{1\times1}{\frac{1}{6}\times0.0821} = \frac{6}{0.0821} = 73.08\ K \simeq 73\ K.$

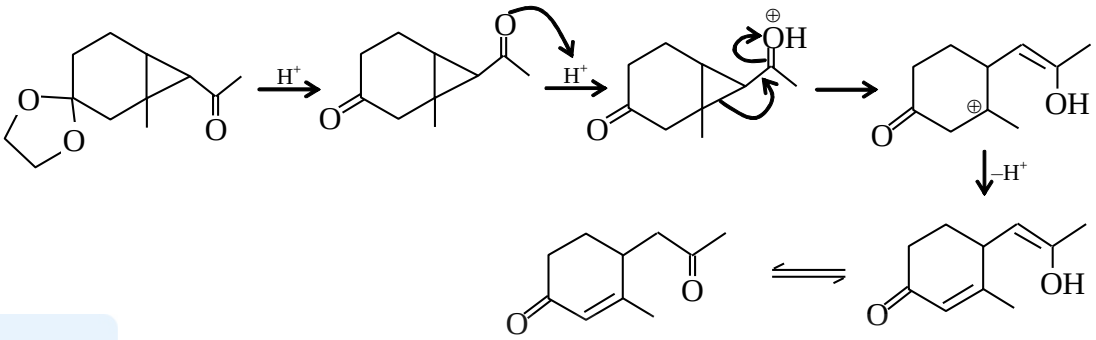
$\nu_{RMS} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3\times8.314\times73}{2\times10^{-3}}} = \sqrt{\mathbf{9.1\times10^5}}\ \mathbf{ms^{-1}}.$

4. **(c)** The second ionization energy of M refers to the reaction, $M^+(g) \longrightarrow M^{2+}(g) + e^-$

$$M(g) \longrightarrow M^{2+}(g) + 2e^- \quad \ldots. \Delta H_5$$
$$M^+(g) + e^- \longrightarrow M(g) \quad \ldots. -\Delta H_3$$
$$M^+(g) \longrightarrow M^{2+}(g) + e^- \quad \Delta H_{IE_2} = \Delta H_5 - \Delta H_3$$

SECTION – II
Multiple Correct Choice Type

5. **(a, d)** The reaction is an example of ring opening of cyclopropane in presence of acid.



6. **(a, b, c)**

$$\underset{10\text{ mL}}{HA} + \underset{10\text{ mL of } 2\times10^{-3}\text{ M}}{NaOH} \longrightarrow NaA + H_2O$$

$$[HA] = \frac{10\times2\times10^{-3}}{10} = 2\times10^{-3}\text{ M}$$

$$\underset{2\times10^{-3}(1-0.2)}{HA(aq)} + H_2O \rightleftharpoons \underset{2\times10^{-3}\times0.2}{A^-(aq)} + \underset{2\times10^{-3}\times0.2}{H_3O^+(aq)}$$

$$K_a = \frac{2\times10^{-3}(0.2)^2}{0.8} = 1\times10^{-4}$$

$$[H_3O^+]_{HA} = 2\times10^{-3}\times0.2 = 4\times10^{-4}$$

$$pH = -\log(4\times10^{-4}) = 4 - \log 4 = 3.4$$

After the neutralization of weak acid (HA) with NaOH, salt (NaA) is formed. NaA gets hydrolysed as

$$\underset{1\times10^{-3}(1-h)}{A^-(aq)} + H_2O \rightleftharpoons \underset{1\times10^{-3}\,h}{HA(aq)} + \underset{1\times10^{-3}\,h}{OH^-(aq)}$$

$$K_h = \frac{K_w}{K_a} = \frac{1\times10^{-14}}{1\times10^{-4}} = 1\times10^{-10}$$

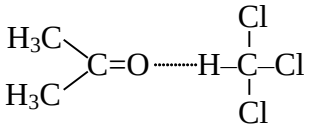
$$K_h = \frac{ch^2}{1-h} \simeq ch^2 \;;\; h = \sqrt{\frac{K_a}{c}} = \sqrt{\frac{1\times10^{-10}}{1\times10^{-3}}}$$

$$[OH^-] = 1\times10^{-3}\times\sqrt{10}\times10^{-4} = \sqrt{10}\times10^{-7}\text{ M}$$

$$pH = -\log\left(\sqrt{10}\times10^{-7}\right) = 6.5$$

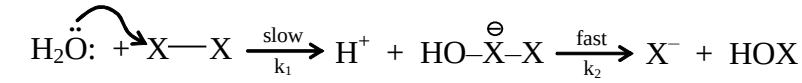
$$pH = 7.5$$

7. **(b, d)**



The mixture of acetone and chloroform show negative deviation because the interaction between acetone and $CHCl_3$ is stronger than that between acetone alone. Pure acetone cannot be obtained from the mixture at azeotropic composition.

8. **(a, c)** The mechanism of the reaction is as follows:



Rate = $k_1[X_2]\,[H_2O] = k_1'[X_2]$

where k_1' is a pseudo first order rate constant with units of sec^{-1} ($time^{-1}$). It is a displacement reaction. H–O–X is not a linear molecule but has a bent shaped structure.

SECTION – III
Reasoning Type

9. **(c)** The product of the reaction of aqueous $FeSO_4$ and NO is $[Fe(H_2O)_5NO]SO_4$, in which Fe is in +1 oxidation state and NO is also in +1 form. It has sp^3d^2 hybridisation.

Paramagnetic species should contain unpaired electron(s) but total number of electrons may be even in the species (for example O_2, S_2).

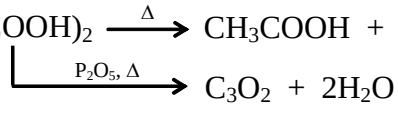
10. **(c)** For monoatomic gas ; $C_v = \frac{3}{2}RT$, $C_p = \frac{5}{2}RT$

For diatomic gas; $C_v = \frac{5}{2}RT$, $C_p = \frac{7}{2}RT$

$$C_v \text{ for mix} = \frac{1\times\frac{3}{2}RT + 1\times\frac{5}{2}RT}{2}$$

$$C_p \text{ for mix} = \frac{1\times\frac{5}{2}RT + 1\times\frac{7}{2}RT}{2} \quad ; \quad \left(\frac{C_p}{C_v}\right)_{mix} = 1.5$$

11. **(c)**



1,3-dioic acid $\xrightarrow[-CO_2]{\Delta}$ Monocarboxylic acid

1,4 and 1,5-dioic acid $\xrightarrow[-H_2O]{\Delta}$ Cyclic anhydride

1,6 and 1,7-dioic acid $\xrightarrow[-CO_2, -H_2O]{\Delta}$ Cyclic ketone

Malonic acid undergoes decarboxylation on heating alone as the reaction proceeds via 6-membered transition state while heating with P_2O_5 promotes loss of H_2O as P_2O_5 is a good dehydrating agent.

12. (a)

SECTION–IV
Comprehension Type

Passage-I

13. (b) $s = [Al^{3+}] + [Al(OH)_4^-]$

$$s = \frac{K_{sp}}{[OH^-]^3} + K_c [OH^-]$$

For minimum solubility, differentiate s w.r.t. $[OH^-]$ and set it equal to zero.

$$\frac{ds}{d[OH^-]} = \frac{-3K_{sp}}{[OH^-]^4} + K_c = 0$$

$$\frac{3K_{sp}}{[OH^-]^4} = K_c$$

$$[OH^-] = \left(\frac{3K_{sp}}{K_c}\right)^{\frac{1}{4}}$$

14. (b) Solubility of $Fe(OH)_3$ decreases with increase in concentration of $[OH^-]$ or decrease in concentration of $[H^+]$.

Addition of further amount of NH_4Cl will increase the concentration of NH_4^+ and thus the ionization of NH_4OH is suppressed. This lowers the $[OH^-]$ concentration and thus pOH increases or pH decreases.

In NH_4I solution, NH_4^+ ion hydrolyses giving H_3O^+, thus making the solution acidic while in HCO_2K solution, HCO_2^- ion hydrolyses to give OH^-, thus solution is basic in nature.

15. (b) $AgCl(s) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq) + Cl^-(aq)$

$$K_1 = K_f \times K_{sp}(AgCl) = 1.6 \times 10^{-10} \times 10^7 = 1.6 \times 10^{-3} = \frac{[(Ag(NH_3)_2^+][Cl^-]}{[NH_3]^2}$$

$AgCN(s) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq) + CN^-(aq)$

$K_2 = K_f \times K_{sp}(AgCN) = 2.5 \times 10^{-16} \times 10^7 = 2.5 \times 10^{-9} = \dfrac{[(Ag(NH_3)_2^+][CN^-]}{[NH_3]^2}$

$\dfrac{[Cl^-]}{[CN^-]} = \dfrac{1.6\times10^{-3}}{2.5\times10^{-9}} = 6.4 \times 10^5$; $K_1 = \dfrac{x^2}{(1-2x)^2}$; $x = [Cl^-] =$ **0.037 M**

$[CN^-] = \dfrac{0.037}{6.4\times10^5} =$ **5.78×10^{-8} M.**

Passage-II

16. **(b)** The optical purity of racemized alcohol $= \dfrac{[\alpha]_{mixture}}{[\alpha]_{pure}} \times 100$

$= \dfrac{-6}{-10} \times 100 =$ **60%**.

17. **(c)** Percentage inversion $= \dfrac{\text{Optical purity of alcohol}}{\text{Optical purity of bromide}} \times 100$

Now, optical purity of bromide $= \dfrac{[\alpha]_{mixture}}{[\alpha]_{pure}} \times 100$

$= \dfrac{30}{36} \times 100 = 83\%.$

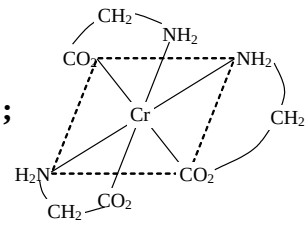
$\therefore$ Percentage inversion $= \dfrac{60}{83} \times 100 =$ **72%**.

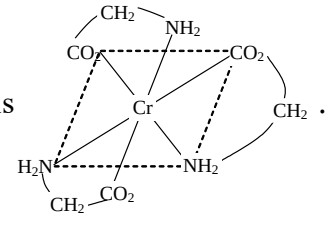
18. **(a)** Total inversion = Pure inverted product + Half of racemized product

$= 72 + \left(\dfrac{1}{2} \times 28\right) = 86\%$

$\therefore$ Percentage of front side attack $= \dfrac{1}{2} \times 28 =$ **14%**.

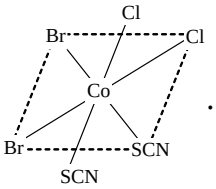
SECTION–IV
Matrix–Match Type

1. **(A) – 2, 3, 4, 5 ;** 

is optically active due to absence of a plane of symmetry and centre of symmetry. It's geometrical isomer is 

.

Cr^{3+} has d^3 configuration, so magnetic moment will be $\sqrt{15}$ and is octahedral in shape.

(B) – 1, 2, 3, 4, 5 ; It can show linkage isomerism due to SCN ligand. Co^{2+} has $3d^7$ configuration, so μ_s will be $\sqrt{15}$. It can show geometrical isomerism and has octahedral geometry. Its optically active isomer will be 

.

(C) – 5 ; Its an octahedral complex.

(D) – 1, 4 ; Linkage isomerism is shown due to NO_2 ligand. It is an square planar complex, which is always optically inactive. It can show geometrical isomerism (3 isomers).

2. **(A) – 1, 4, 5 ;** (A) has a chiral centre and can show geometrical isomerism. Stereogenic units are structural units around which stereoisomers can be generated.

(B) – 2, 3, 5 ; (B) is optically inactive due to presence of plane of symmetry and it has 2 stereogenic units.

(C) – 1, 4 ; (C) is optically active as it has one chiral centre.

(D) – 2, 3, 5 ; (D) is optically inactive due to symmetry (plane of symmetry) and possesses two stereogenic units.

mathongo